

Introduction to Image Processing

Wigner Summer Camp

Data and Compute Intensive Sciences Research Group

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Outline

Introduction

Pipeline

Methods

Applications

Project Overview

- ▶ Image processing techniques for extracting useful information
- ▶ Filtering, transformations and feature extraction
- ▶ Python based implementation
- ▶ Goal: analyze and enhance digital images

What is Image Processing?

Definition

Image processing is the manipulation and analysis of digital images to improve quality or extract important information.

- ▶ Image enhancement
- ▶ Noise reduction
- ▶ Edge detection
- ▶ Feature extraction

Outline

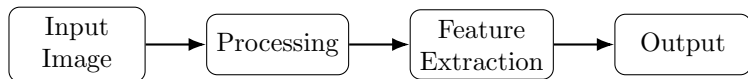
Introduction

Pipeline

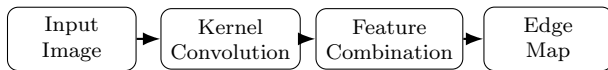
Methods

Applications

Main Pipeline



Architecture



- ▶ Image passes through convolution operations
- ▶ Features are extracted
- ▶ Output represents important structures

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Core Methods

Filtering

- ▶ Blur
- ▶ Sharpening
- ▶ Noise removal

Detection

- ▶ Edge detection
- ▶ Feature extraction
- ▶ Image analysis

Technology Stack

Language	Python
Libraries	OpenCV, NumPy, Matplotlib
Input	Digital images
Output	Processed images

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Applications

- ▶ Computer vision systems
- ▶ Object detection
- ▶ Medical image analysis
- ▶ Autonomous systems
- ▶ Industrial inspection

Summary

Conclusion

Image processing allows computers to analyze and manipulate visual information using mathematical methods.

- ▶ Transform images
- ▶ Extract features
- ▶ Detect patterns